

Optimization

2.4 Bisection method

José Firmino Aguilar Madeira

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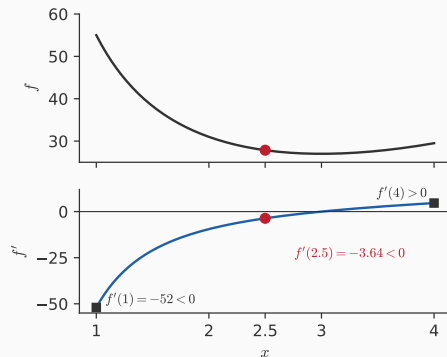
With values of f only (2.1–2.2) we needed **two** points to eliminate. With f' : at an interior minimum, $f'(x^*) = 0$.

Can we do the same using only the sign of f' ?

Guiding example: $f(x) = x^2 + 54/x$ on $[1, 4]$,

$f'(x) = 2x - 54/x^2$. A single point:

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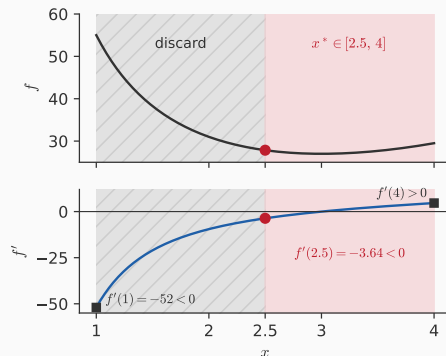
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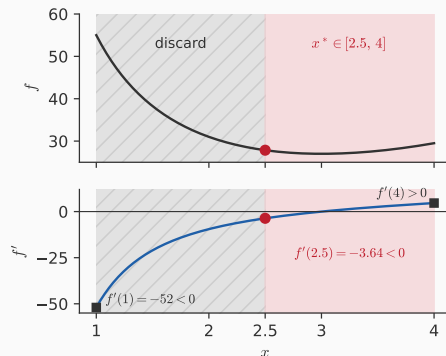
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Where to put the point? **In the middle**: whatever the sign, half remains.

sign of f' at the midpoint $\rightarrow$ **bisection**



Why the sign is enough

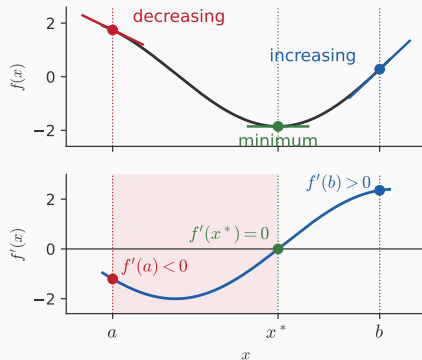
Assumptions: f' continuous on $[a, b]$,
 $f'(a) < 0 < f'(b)$ and **a single zero**, where f'
changes sign $- \rightarrow +$. Then that zero is the only
interior minimum x^* . In particular, f strictly
convex with $f'(a) < 0 < f'(b)$ is enough.

The sign of f' at a point tells **on which side** the
minimum is:

- $f'(x) < 0$: f is still decreasing $\Rightarrow x^* > x$;
- $f'(x) > 0$: f is already increasing $\Rightarrow x^* < x$.

It tells *on which side*, not *where*:

$f' = -3.64$ and $f' = -0.0001$ give the same
decision.



$f'(a) < 0 < f'(b)$ alone guarantees *at least one*
zero; without uniqueness there may be several
stationary points and we do not know whether we
isolate the desired minimum.

The algorithm

Bisection: $[a, b]$ with $f'(a) < 0 < f'(b)$,

ε_x

```
1: set  $k = 0$ ;  $a_0 = a$ ;  $b_0 = b$ 
2: while  $b_k - a_k > \varepsilon_x$  do
3:   set  $x_k = \frac{a_k + b_k}{2}$ 
4:   if  $f'(x_k) = 0$  then return  $x_k$ 
5:   end if
6:   if  $f'(x_k) < 0$  then  $\triangleright$  minimum to the right
7:     set  $a_{k+1} = x_k$ ;  $b_{k+1} = b_k$ 
8:   else  $\triangleright$  minimum to the left
9:     set  $a_{k+1} = a_k$ ;  $b_{k+1} = x_k$ 
10:  end if
11:  set  $k = k + 1$ 
12: end while
13: return  $x^* \approx (a_k + b_k)/2$ 
```

- **Order 1** information: it uses f' (only its sign!), not f .
- It requires $f'(a) < 0 < f'(b)$. If this condition fails, the minimum is not bracketed under the method's conditions and bisection should not be applied in this form — check before starting.
- If $f'(x_k) = 0$ exactly, x_k is the minimum: stop (in finite arithmetic this is rare).
- Cost: **one evaluation of f' per iteration**. If f' is not available analytically, it is approximated by finite differences (2.6) — two evaluations of f .

How many iterations? And how much does each one cost?

Each iteration halves the interval:

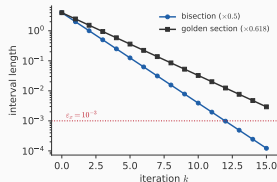
$b_k - a_k = \left(\frac{1}{2}\right)^k (b_0 - a_0)$. With ε_x the tolerance on the interval length (as in the `while`), $b_k - a_k \leq \varepsilon_x$ — and hence $|x - x^*| \leq \varepsilon_x/2$ at the midpoint — when

$$k = \left\lceil \log_2 \frac{b_0 - a_0}{\varepsilon_x} \right\rceil$$

Example

$[a_0, b_0] = [0, 2]$, $\varepsilon_x = 10^{-3}$: $\log_2 2000 = 10.97$, so $k = 11$. With $\varepsilon_x = 10^{-4}$: $k = 15$.

Each decimal digit costs $\log_2 10 \approx 3.3$ iterations (golden section: 4.8).



But what is the cost per evaluation of f ?

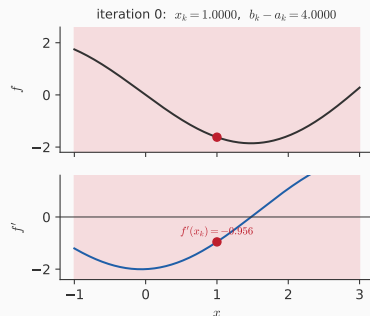
Bisection: $\times 0.5$ per f' ; golden section: $\times 0.618$ per new f . With f' by central differences (2 evaluations of f): $\sqrt{0.5} \approx 0.707$ per evaluation — **worse**.

A cost conversion, not an intrinsic rate of bisection; it assumes 2 new evaluations of f per f' , with no reuse.

Example 1: $f(x) = -2 \sin x + x^2/16$ on $[-1, 3]$

$$f'(x) = -2 \cos x + x/8; \quad f'(-1) = -1.21 < 0, \\ f'(3) = 2.35 > 0 \quad \checkmark$$

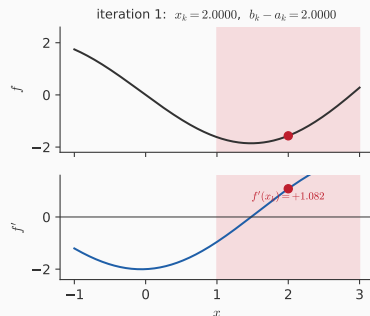
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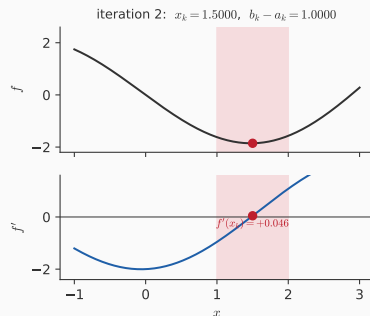
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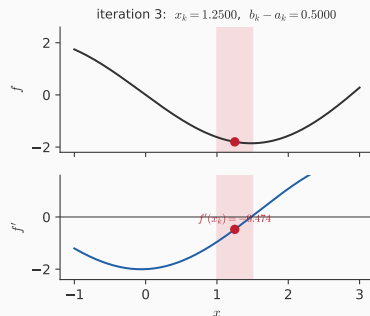
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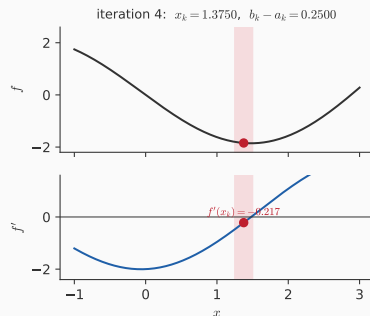
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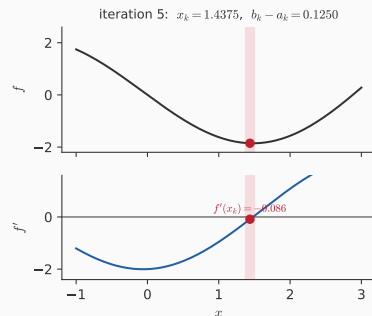
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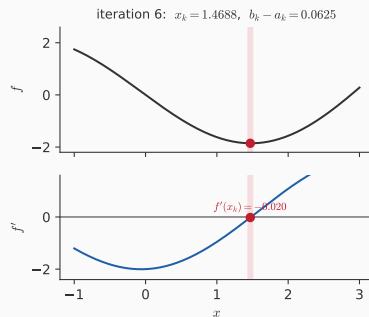
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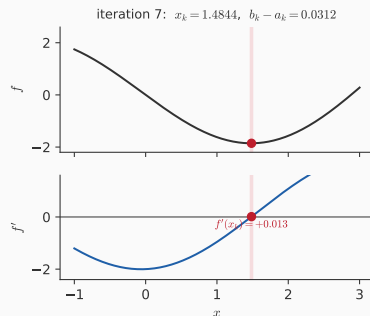


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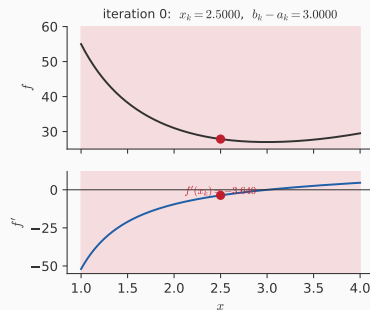
After 15 reductions, midpoint: $x^* \approx 1.4783$, $f(x^*) \approx -1.8549$,
 $b - a = 1.2 \times 10^{-4}$.



Example 2: $f(x) = x^2 + 54/x$ on $[1, 4]$

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(same interval as golden section in 2.2)

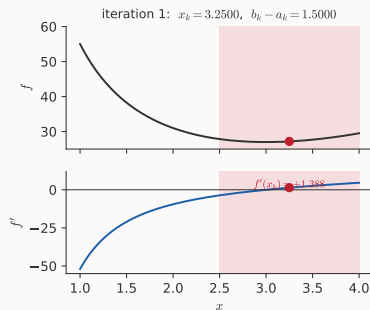
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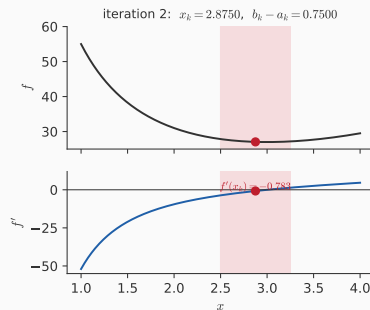
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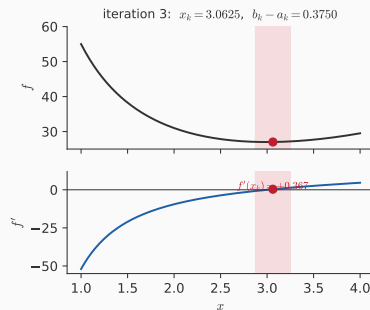
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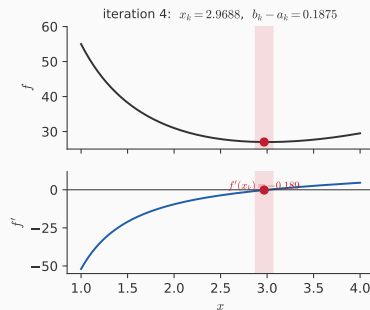
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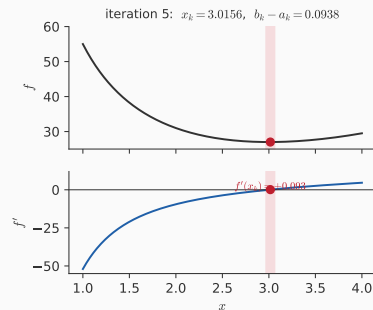
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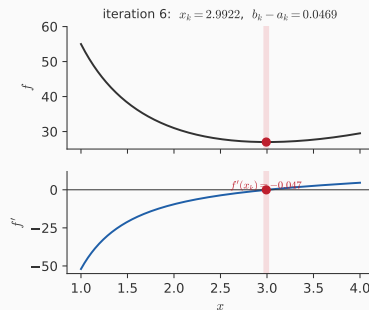
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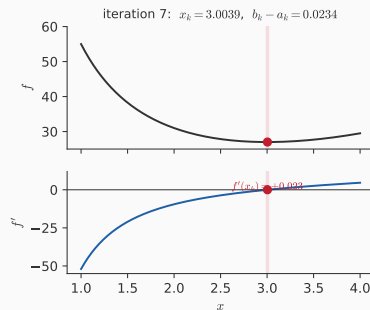


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After 15 reductions, midpoint: $x^* \approx 3.0000$,
 $b - a = 3/2^{15} = 9.2 \times 10^{-5}$ (exact: $x^* = 3$). Golden section, same
 $[1, 4]$, 15 reductions: $L = 0.0022$, 24 times larger (but with f only).
The $f(x_k)$ column is for reading only: the method does not use it.



To solve

Apply bisection with 15 reductions (final midpoint) and confirm:

$f(x)$	interval	x^*	$f(x^*)$
$x + \frac{31}{x+3}$	$[0, 5]$	2.5678	8.1355
$-3 \sin x + \frac{x^2}{8}$	$[0, 4]$	1.4496	-2.7153

- In the first, check analytically: $f'(x) = 1 - \frac{31}{(x+3)^2} = 0 \Rightarrow x^* = \sqrt{31} - 3 = 2.5678$.
- In the second, compare with golden section in 2.2 (1.4489, $L_{15} = 0.0029$) and with bisection ($b - a = 4/2^{15} = 1.2 \times 10^{-4}$): after 15 reductions, which one guarantees the smaller error?

On the computer

The core, in MATLAB (N reductions):

```
if ~(df(a)<0 && df(b)>0), error('bracketing'), end
for k = 1:N
    x = (a + b)/2; d = df(x);
    if d == 0
        a = x; b = x; break % finished
    elseif d < 0
        a = x; % minimum to the right
    else
        b = x; % minimum to the left
    end
end
x = (a + b)/2;
```

The full function (Bisection) is in the course companion code.

It uses only f' : $n_g = 1$ per reduction, plus $f'(a)$ and $f'(b)$ to check the bracketing ($n_g = N + 2$); $n_f = 0$.

$f'(x_k) = 0$: **finished** ($x_k = x^*$). In finite arithmetic an exact zero is rare; the main criterion is the interval length.

Without an analytical f' , central differences (2.6), $n_f = 2$ per f' :

```
h = 1e-6;
df = @(x) (f(x+h) - f(x-h)) / (2*h);
```

near the zero, the **sign** of f' may also have numerical noise — see 2.6

KEY TAKEAWAYS — 2.4

1. For an interior minimum of a differentiable f , $f'(x^*) = 0$; under the assumptions of this section, minimizing is **equivalent** to locating that zero.
2. With f' continuous, $f'(a) < 0 < f'(b)$ and a single zero, the **sign** at the midpoint tells which half to keep: $f'(x_k) < 0 \Rightarrow$ right; $f'(x_k) > 0 \Rightarrow$ left; $f'(x_k) = 0 \Rightarrow$ finished.
3. $L_k = 2^{-k}L_0$: linear convergence with ratio $\frac{1}{2}$ and a predictable number of reductions, $k = \lceil \log_2(L_0/\varepsilon_x) \rceil$. One evaluation of f' per reduction, none of f (with f' by central differences, golden section reduces more per evaluation of f).
4. It uses only the sign of f' : **it knows on which side the zero is, but not where.**

Next (2.5): what if we also use the *value* of $f'(x_k)$ and the slope of f' , that is $f''(x_k)$, to **estimate directly where** the zero is? — the Newton–Raphson method.

References and resources

References Full list: *Course bibliography*, at the end of the course PDF.

- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012 — Ch. 2.
- Dennis, J. E., Jr., Schnabel, R. B., *Numerical Methods for Unconstrained Optimization and Nonlinear Equations*, SIAM (Classics in Applied Mathematics 16), 1996 — Ch. 2.

Resources

- Companion code: Bisection; f' by central differences (df, $h = 10^{-6}$).